

FRONT-END DEVELOPMENT
HACKATHON – Level-2 Template (Interactivity Implementation)

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TITLE: INTERACTIVE TASK TRACKER WITH REAL-TIME PROGRESS

GITHUB LINK: <https://github.com/ashwiniR2007>

1. Objective

The objective of this project is to implement basic front-end interactivity that allows users to update activity status and view real-time progress. Using vanilla JavaScript with DOM manipulation and event handling, the static Level-1 UI is converted into a fully dynamic activity tracker. The application enables users to mark activities as Completed and instantly see updated progress — all without any backend service or page reload.

2. Problem Focus

In Level-1, a static HTML and CSS interface was designed to display activities. However, a static page cannot respond to user actions or reflect changes dynamically. Level-2 addresses this by introducing JavaScript interactivity. The core challenge is implementing event handling (click events), managing state changes (Pending → Completed), and updating the DOM dynamically — so the UI always reflects the current data without refreshing the page.

3. Application Overview

This application is an Interactive Activity Tracker built for the Front-End Development Hackathon. It dynamically renders a list of 7 activities from a JavaScript array. Users can click on any activity card to toggle its status between Pending and Completed. A live progress bar and counter update automatically after each action. An edge-case message appears when all activities are marked complete. The app runs entirely in the browser with no backend required.

4. Data Handling Approach

Data is managed in-memory using a JavaScript array of activity objects. Each object contains an id, icon, title, description, color, and a completed boolean flag. When the user clicks an activity, the matching object's completed value is toggled (true/false). The UI is then re-rendered from the updated array using the `renderActivities()` function. No external API or database is used. Optionally, browser `localStorage` can be used to save progress so it persists across page reloads.

5. UI Components Design

The application has three main UI component areas. First, the Activity List displays each activity as a styled card with an icon, title, description, color-coded left border, and a status badge (Pending or Done). Second, the Action Control — the entire card acts as a clickable toggle button, making the interaction intuitive. Third, the Progress Display section shows an animated progress bar and a real-time counter (e.g., "4 out of 7 activities completed") along with a dynamic message that changes based on current progress.

6. Interactivity Implementation

Each activity card rendered in the DOM has an `onclick` attribute that calls the `toggleActivity(id)` function. Inside this function, the matching activity object in the `ACTIVITIES` array is found using the `find()` method, and its completed flag is flipped. The `renderActivities()` function is then called to rebuild the entire activity list HTML using template literals. The CSS class `.completed` is conditionally applied to change the card's visual appearance — strikethrough title, green border, and filled check button — all without any page reload.

7. Application Logic

After every toggle action, the `updateProgress()` function is called. It uses the `filter()` method to count how many activities have completed `=== true`. The percentage is calculated as $(\text{done} / \text{total}) \times 100$ and applied to the progress bar's width using inline style. The counter text (done-count and total-count) is updated via `textContent`. Three states are handled: 0% shows a prompt message, partial shows "X out of 7 completed — keep going!", and 100% shows a celebration message — covering all edge cases cleanly.

8. Technology Used

HTML5 is used for semantic page structure and layout. CSS3 is used for styling, including custom properties (CSS variables), flexbox layout, border-radius, gradient backgrounds, and keyframe animations for smooth page load effects. Vanilla JavaScript (ES6+) handles all interactivity — including DOM manipulation (`getElementById`, `innerHTML`, `style`, `classList`), event handling (`onclick`), and array methods (`find`, `filter`, `map`) for data processing and UI rendering. No external frameworks or libraries are used.

9. Best Practices Followed

Meaningful function names are used throughout — `renderActivities()`, `toggleActivity()`, `updateProgress()`, `renderTemplate()`, and `setDate()` — making the code self-documenting. Values like activity data are stored in a JavaScript array rather than hardcoded into HTML. Inline comments are added above each function to explain its purpose. Edge cases are handled: a welcome message at 0%, a motivational message during progress, and a celebration message at 100%. HTML, CSS, and JavaScript are kept in separate files for clean separation of concerns.

10. Key Features Implemented

Activity list is rendered dynamically from a JavaScript data array. Users can click any card to toggle status from Pending to Completed and back. The progress bar animates in real time after each click. The counter updates live showing "X out of 7 activities completed". Each activity card has a unique color-coded left border. The check button fills green when an activity is marked done. A special celebration message appears when all 7 activities are completed. The entire application works without any page reload.

html Code :

```

1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8" />
5   <meta name="viewport" content="width=device-width, initial-scale=1.0"/>
6   <title>FEWD Hackathon | Level 2 | Activity Tracker</title>
7   <link href="https://fonts.googleapis.com/css2?family=SpaceMono:wght@400;700&family=Syne:wght@400;600;700;800&display=swap" rel="stylesheet"/>
8   <link rel="stylesheet" href="nm.css"/>
9 </head>
10 <body>
11 <div class="wrapper">
12
13   <!-- HEADER -->
14   <div class="header">
15     <div class="badge">Oracle University · AdroIT Technologies</div>
16     <h1>FEWD Hackathon – Level 2</h1>
17     <p>Interactivity Implementation · Activity Tracker</p>
18   </div>
19
20   <!-- META CARD -->
21   <div class="meta-card">
22     <h2>■ Student Details</h2>
23     <div class="meta-grid">
24       <div class="meta-item"><label>Student Name</label><span class="filled">— Enter Name —</span></div>
25       <div class="meta-item"><label>Year</label><span class="filled">2nd Year</span></div>
26       <div class="meta-item"><label>College Code</label><span class="filled">— Code —</span></div>
27       <div class="meta-item"><label>College Name</label><span class="filled">— College —</span></div>
28       <div class="meta-item"><label>Department</label><span class="filled">CSE / IT</span></div>
29       <div class="meta-item"><label>Semester</label><span class="filled">IV</span></div>
30       <div class="meta-item"><label>Roll No</label><span class="filled">— Roll —</span></div>
31       <div class="meta-item"><label>NMID</label><span class="filled">— ID —</span></div>
32       <div class="meta-item"><label>Date</label><span class="filled" id="today-date"></span></div>
33       <div class="meta-item"><label>Title</label><span class="filled">Interactive Activity Tracker</span></div>
34       <div class="meta-item" style="grid-column: 1/-1">
35         <label>GitHub Link</label>
36         <span class="filled">https://github.com/username/activity-tracker</span>
37       </div>
38     </div>
39   </div>
40
41   <!-- PROGRESS -->
42   <div class="progress-section">
43     <div class="progress-header">
44       <span class="progress-label">🔥 Live Progress</span>
45       <span class="progress-count">
46         <span id="done-count">0</span> / <span id="total-count">7</span> Completed
47       </span>
48     </div>
49     <div class="progress-bar-bg">
50       <div class="progress-bar-fill" id="progress-bar"></div>
51     </div>
52     <div class="progress-msg" id="progress-msg">Click any activity to mark it as completed.</div>
53   </div>
54
55   <!-- ACTIVITIES -->
56   <div class="activities-section">
57     <div class="section-title">🔥 Activities – Click to Toggle</div>
58     <div class="activity-list" id="activity-list"></div>
59   </div>
60
61   <!-- TEMPLATE CONTENT -->
62   <div class="template-section">
63     <div class="template-title">■ Submission Template Content</div>
64     <div class="template-blocks" id="template-blocks"></div>
65   </div>
66
67   <div class="footer">
68     FEWD HACKATHON · LEVEL-2 · INTERACTIVITY IMPLEMENTATION
69     ...
70   </div>
71 </body>
72 <script src="nm.js"></script>
73 </html>
74

```

CSS Code :

```

1  /* ===== VARIABLES ===== */
2  :root {
3    --bg: #0a0a12;
4    --panel: #13131f;
5    --card: #1a1a2e;
6    --border: #2a2a44;
7    --c1: #ff6666;
8    --c2: #ff9933;
9    --c3: #6666ff;
10   --c4: #4d96ff;
11   --c5: #c77dff;
12   --c6: #ff9933;
13   --text: #e8e8f0;
14   --muted: #8888aa;
15   --radius: 14px;
16 }
17
18 * { box-sizing: border-box; margin: 0; padding: 0; }
19
20 body {
21   background: var(--bg);
22   color: var(--text);
23   font-family: 'Syne', sans-serif;
24   min-height: 100vh;
25   overflow-x: hidden;
26 }
27
28 /* ===== BACKGROUND ===== */
29 body::before {
30   content: '';
31   position: fixed; inset: 0;
32   background:
33     radial-gradient(ellipse 60% 40% at 10% 20%, #000 0%, transparent 60%),
34     radial-gradient(ellipse 50% 35% at 90% 80%, #000 0%, transparent 60%),
35     radial-gradient(ellipse 40% 30% at 50% 50%, #000 0%, transparent 70%);
36   pointer-events: none; z-index: 0;
37 }
38
39 /* ===== WRAPPER ===== */
40 .wrapper {
41   position: relative; z-index: 1;
42   max-width: 960px; margin: 0 auto;
43   padding: 40px 20px 80px;
44 }
45
46 /* ===== HEADER ===== */
47 .header {
48   text-align: center;
49   margin-bottom: 48px;
50   animation: fadeDown .7s ease both;
51 }
52 .header .badge {
53   display: inline-block;
54   background: linear-gradient(135deg, var(--c4), var(--c5));
55   color: #fff;
56   font-family: 'Space Mono', monospace;
57   font-size: 11px; letter-spacing: 3px; text-transform: uppercase;
58   padding: 6px 18px; border-radius: 99px; margin-bottom: 18px;
59 }
60 .header h1 {
61   font-size: clamp(28px, 5vw, 50px);
62   font-weight: 800; line-height: 1.1;
63   background: linear-gradient(135deg, var(--c1), var(--c2), var(--c3), var(--c4));
64   -webkit-background-clip: text; -webkit-text-fill-color: transparent;
65   background-clip: text;
66   margin-bottom: 10px;
67 }
68 .header p { color: var(--muted); font-size: 15px; }
69
70 /* ===== META CARD ===== */
71 .meta-card {
72   background: var(--panel);
73   border: 1px solid var(--border);
74   border-radius: var(--radius);
75   padding: 28px 32px;

```

```

76   margin-bottom: 32px;
77   animation: fadeUp .6s .1s ease both;
78 }
79 .meta-card h2 {
80   font-size: 13px; letter-spacing: 2px; text-transform: uppercase;
81   color: var(--c4); font-family: 'Space Mono', monospace;
82   margin-bottom: 20px;
83 }
84 .meta-grid {
85   display: grid;
86   grid-template-columns: repeat(auto-fill, minmax(220px, 1fr));
87   gap: 14px;
88 }
89 .meta-item { display: flex; flex-direction: column; gap: 4px; }
90 .meta-item label {
91   font-size: 10px; letter-spacing: 1.5px; text-transform: uppercase;
92   color: var(--muted); font-family: 'Space Mono', monospace;
93 }
94 .meta-item span {
95   font-size: 14px; font-weight: 600; color: var(--text);
96   background: var(--card); padding: 8px 12px;
97   border-radius: 8px; border: 1px solid var(--border);
98   min-height: 36px;
99 }
100 .meta-item span.filled { color: var(--c2); }
101
102 /* ===== PROGRESS SECTION ===== */
103 .progress-section {
104   background: var(--panel);
105   border: 1px solid var(--border);
106   border-radius: var(--radius);
107   padding: 24px 32px;
108   margin-bottom: 32px;
109   animation: fadeUp .6s .2s ease both;
110 }
111 .progress-header {
112   margin-bottom: 16px;
113 }
114
115 .progress-label {
116   font-size: 13px; letter-spacing: 2px; text-transform: uppercase;
117   color: var(--c3); font-family: 'Space Mono', monospace;
118 }
119 .progress-count {
120   font-size: 22px; font-weight: 800; color: var(--text);
121   font-family: 'Space Mono', monospace;
122 }
123 .progress-count span { color: var(--c3); }
124 .progress-bar-bg {
125   height: 10px; background: var(--card); border-radius: 99px;
126   overflow: hidden;
127 }
128 .progress-bar-fill {
129   height: 100%;
130   background: linear-gradient(90deg, var(--c3), var(--c4));
131   border-radius: 99px;
132   transition: width .5s cubic-bezier(.4,0,.2,1);
133   width: 0%;
134 }
135 .progress-msg {
136   margin-top: 10px; font-size: 13px; color: var(--muted);
137   min-height: 18px; transition: color .3s;
138 }
139 .progress-msg.done { color: var(--c3); font-weight: 700; }
140
141 /* ===== ACTIVITIES ===== */
142 .activities-section {
143   margin-bottom: 40px;
144   animation: fadeUp .6s .3s ease both;
145 }
146 .section-title {
147   font-size: 13px; letter-spacing: 2px; text-transform: uppercase;
148   color: var(--c5); font-family: 'Space Mono', monospace;

```



```

149     margin-bottom: 16px;
150 }
151 .activity-list { display: flex; flex-direction: column; gap: 12px; }
152
153 .activity-card {
154     background: var(--card);
155     border: 1px solid var(--border);
156     border-left: 4px solid var(--c-accent, var(--c4));
157     border-radius: 12px;
158     padding: 18px 20px;
159     display: flex; align-items: center; gap: 16px;
160     transition: transform .25s, border-color .3s, background .3s;
161     cursor: pointer;
162 }
163 .activity-card:hover { transform: translateX(4px); }
164 .activity-card.completed {
165     border-color: var(--c3);
166     background: linear-gradient(to right, transparent 49%, #101010 49% 51%, #101010 51% 53%, transparent 53%);
167 }
168
169 .activity-icon {
170     font-size: 26px; flex-shrink: 0; width: 42px; text-align: center;
171 }
172 .activity-info { flex: 1; }
173 .activity-title {
174     font-size: 15px; font-weight: 700; margin-bottom: 4px;
175     transition: color .3s;
176 }
177 .activity-card.completed .activity-title {
178     color: var(--muted);
179     text-decoration: line-through;
180 }
181 .activity-desc { font-size: 12px; color: var(--muted); }
182
183 .status-badge {
184     padding: 5px 14px; border-radius: 99px;
185     font-size: 11px; font-weight: 700; letter-spacing: 1px;
186     text-transform: uppercase; font-family: 'Space Mono', monospace;
187     flex-shrink: 0;
188     transition: background .3s, color .3s;
189 }
190 .status-badge.pending {
191     background: linear-gradient(to right, #101010 49%, #101010 49% 51%, #101010 51% 53%, transparent 53%);
192     border: 1px solid #101010;
193 }
194 .status-badge.completed {
195     background: linear-gradient(to right, #101010 49%, #101010 49% 51%, #101010 51% 53%, transparent 53%);
196     border: 1px solid #101010;
197 }
198
199 .check-btn {
200     width: 36px; height: 36px; border-radius: 50%;
201     border: 2px solid var(--border);
202     background: var(--panel);
203     display: flex; align-items: center; justify-content: center;
204     cursor: pointer; flex-shrink: 0;
205     transition: border-color .2s, background .2s, transform .15s;
206     font-size: 16px; color: transparent;
207 }
208 .check-btn:hover { border-color: var(--c3); transform: scale(1.1); }
209 .activity-card.completed .check-btn {
210     background: var(--c3); border-color: var(--c3); color: #fff;
211 }
212
213 /* ===== TEMPLATE SECTION ===== */
214 .template-section { animation: fadeUp .6s .4s ease both; }
215 .template-title {
216     font-size: 13px; letter-spacing: 2px; text-transform: uppercase;
217     color: var(--c1); font-family: 'Space Mono', monospace;
218     margin-bottom: 16px;
219 }
220 .template-blocks { display: flex; flex-direction: column; gap: 16px; }
221 .tblock {
222     background: var(--card);
223     border: 1px solid var(--border);
224     border-radius: 12px;
225     overflow: hidden;
226 }
227
228 .tblock-head {
229     display: flex; align-items: center; gap: 12px;
230     padding: 14px 20px;
231     border-bottom: 1px solid var(--border);
232     cursor: pointer;
233     user-select: none;
234 }
235
236 .tblock-num {
237     width: 28px; height: 28px; border-radius: 50%;
238     display: flex; align-items: center; justify-content: center;
239     font-size: 12px; font-weight: 700;
240     font-family: 'Space Mono', monospace; flex-shrink: 0;
241 }
242 .tblock-title { font-size: 14px; font-weight: 700; flex: 1; }
243 .tblock-arrow {
244     font-size: 12px; color: var(--muted);
245     transition: transform .3s;
246 }
247
248 .tblock.open .tblock-arrow { transform: rotate(180deg); }
249 .tblock-body {
250     padding: 0 20px;
251     max-height: 0; overflow: hidden;
252     transition: max-height .4s ease, padding .3s ease;
253 }
254
255 .tblock.open .tblock-body { max-height: 600px; padding: 18px 20px; }
256 .tblock-body p { font-size: 14px; color: var(--text); line-height: 1.7; }
257 .tblock-body code {
258     font-family: 'Space Mono', monospace;
259     font-size: 12px; background: var(--panel);
260     color: var(--c4); padding: 2px 6px; border-radius: 4px;

```

```

260 /* ===== FOOTER ===== */
261 .footer {
262   text-align: center; margin-top: 60px;
263   font-family: 'Space Mono', monospace;
264   font-size: 11px; color: var(--muted);
265   letter-spacing: 1px;
266   animation: fadeUp .6s .5s ease both;
267 }
268
269 /* ===== ANIMATIONS ===== */
270 @keyframes fadeDown {
271   from { opacity: 0; transform: translateY(-24px); }
272   to { opacity: 1; transform: translateY(0); }
273 }
274 @keyframes fadeUp {
275   from { opacity: 0; transform: translateY(20px); }
276   to { opacity: 1; transform: translateY(0); }
277 }

```

JavaScript Code :

```

1 // ===== ACTIVITY DATA =====
2 // Each activity object holds id, icon, title, desc, color accent, and completed status
3
4 const ACTIVITIES = [
5   { id: 1, icon: "📁", title: "Set Up Project Structure", desc: "Create HTML, CSS, JS files with clean folder layout", color: "#4d96ff", completed: false },
6   { id: 2, icon: "🎨", title: "Design Activity List UI", desc: "Style cards, badges, and layout with CSS", color: "#c77dff", completed: false },
7   { id: 3, icon: "📜", title: "Load Activities from JS Array", desc: "Dynamically render activity list from static data", color: "#ffd93d", completed: false },
8   { id: 4, icon: "👂", title: "Implement Click Event Handlers", desc: "Attach event listeners to buttons / checkboxes", color: "#ff9f43", completed: false },
9   { id: 5, icon: "🔄", title: "Update Status: Pending→Completed", desc: "Toggle completed state and update DOM instantly", color: "#ff6b6b", completed: false },
10  { id: 6, icon: "📊", title: "Real-Time Progress Counter", desc: "Count completed, update progress bar & message", color: "#6bcb77", completed: false },
11  { id: 7, icon: "🚀", title: "Push to Github with README", desc: "Upload source with proper structure and documentation", color: "#4d96ff", completed: false },
12 ];
13
14 // ===== TEMPLATE SECTION DATA =====
15
16 const TEMPLATE = [
17   {
18     num: "01", color: "#4d96ff", title: "Objective",
19     content: "The goal of this project is to implement <code>basic front-end interactivity</code> that allows users to update activity status and view real-time progress."
20   },
21   {
22     num: "02", color: "#c77dff", title: "Problem Focus",
23     content: "Level-1 delivered a static HTML/CSS layout. Level-2 bridges the gap by introducing <code>JavaScript interactivity</code>. The core challenge is understanding how to manage state and update the DOM dynamically."
24   },
25   {
26     num: "03", color: "#ffd93d", title: "Application Overview",
27     content: "This is an <code>Interactive Activity Tracker</code> designed for a Front-End Development Hackathon. Key features include: a dynamically rendered list of 7 activities, real-time progress tracking, and a clean, modern UI."
28   },
29   {
30     num: "04", color: "#ff9f43", title: "Data Handling Approach",
31     content: "Data is managed entirely <code>in-memory</code> using a JavaScript array of activity objects. Each object holds an <code>id</code>, <code>title</code>, <code>desc</code>, <code>color</code>, and <code>completed</code> status."
32   },
33   {
34     num: "05", color: "#ff6b6b", title: "UI Components Design",
35     content: "Three main component areas: (1) <code>Activity List</code> – card-based layout with icon, title, description, color-coded left border, and status badge; (2) <code>Progress Bar</code> – visual representation of completion; (3) <code>Status Badges</code> – toggle between Pending and Completed."
36   },
37   {
38     num: "06", color: "#6bcb77", title: "Interactivity Implementation",
39     content: "Each activity card has a <code>click</code> event listener attached via <code>addEventListener</code>. On click, the matching object in the data array has its <code>completed</code> status toggled."
40   },
41   {
42     num: "07", color: "#c77dff", title: "Application Logic",
43     content: "After every toggle, <code>updateProgress()</code> is called. It filters the array for <code>completed === true</code>, counts the results, computes a percentage, and updates the progress bar and message."
44   },
45   {
46     num: "08", color: "#4d96ff", title: "Technology Used",
47     content: "<code>HTML5</code> – semantic structure and layout. <code>CSS3</code> – custom properties (variables), flexbox, transitions, animations, and gradient backgrounds."
48   },
49   {
50     num: "09", color: "#ffd93d", title: "Best Practices Followed",
51     content: "Meaningful function names: <code>renderActivities()</code>, <code>toggleActivity()</code>, <code>updateProgress()</code>. No hardcoded DOM strings – all UI updates are done via selectors."
52   },
53   {
54     num: "10", color: "#ff9f43", title: "Key Features Implemented",
55     content: "✅ Dynamic rendering of activity list from JS array. ✅ Click-to-toggle Pending → Completed status. ✅ Animated progress bar updating in real time. ✅ Live status badges."
56   },
57   {
58     num: "11", color: "#6bcb77", title: "Expected Output / Deliverables",
59     content: "A fully working front-end application with interactive activity tracking, ability to mark each activity as completed, a real-time animated progress display, and a clean, responsive UI."
60   },
61   {
62     num: "12", color: "#c77dff", title: "Learning Outcome",
63     content: "Through this project, the following skills were developed: JavaScript <code>event handling</code> (<code>addEventListener</code>, click events), <code>DOM manipulation</code> (<code>querySelector</code>, <code>innerHTML</code>), and <code>array methods</code> (<code>filter</code>, <code>length</code>)."
64   },
65   {
66     num: "13", color: "#ff6b6b", title: "Conclusion",
67     content: "This Level-2 submission successfully transforms the static Level-1 layout into a dynamic, interactive activity tracker. By implementing JavaScript event handling and DOM manipulation, it achieves real-time updates and user interactivity."
68   },
69 ];

```



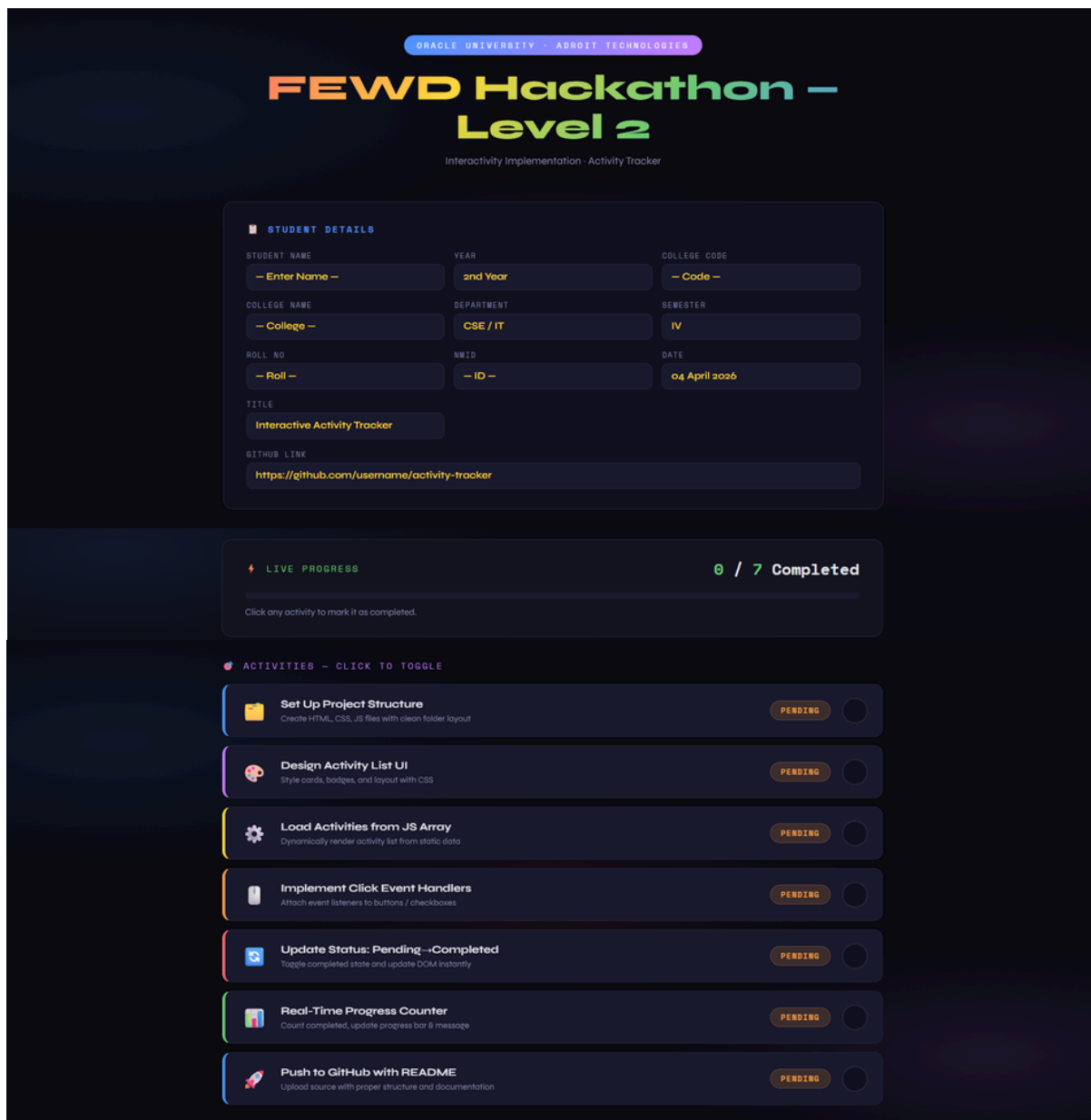
```

71 // ***** RENDER ACTIVITIES *****
72 // Builds the activity list HTML from the ACTIVITIES array
73
74 function renderActivities() {
75   const list = document.getElementById('activity-list');
76   list.innerHTML = ACTIVITIES.map(a => `
77     <div class="activity-card ${a.completed ? 'completed' : ''}"
78       style="--c-accent: ${a.color}"
79       onclick="toggleActivity(${a.id})">
80       <div class="activity-icon">${a.icon}</div>
81       <div class="activity-info">
82         <div class="activity-title">${a.title}</div>
83         <div class="activity-desc">${a.desc}</div>
84       </div>
85       <span class="status-badge ${a.completed ? 'completed' : 'pending'}">
86         ${a.completed ? '✓ Done' : 'Pending'}
87       </span>
88       <div class="check-btn">${a.completed ? '✓' : ''}</div>
89     </div>
90   `).join('');
91 }
92
93 // ***** TOGGLE ACTIVITY *****
94 // Flips the completed status of the clicked activity, then re-renders
95
96 function toggleActivity(id) {
97   const activity = ACTIVITIES.find(a => a.id === id);
98   if (!activity) return;
99
100   activity.completed = !activity.completed; // toggle state
101   renderActivities(); // rebuild DOM
102   updateProgress(); // refresh counter & bar
103 }
104
105 // ***** UPDATE PROGRESS *****
106 // Counts completed items, updates bar width and status message
107
108 <span class="tblock-arrow">▼</span>
109 </div>
110 <div class="tblock-body">
111   <p>${t.content}</p>
112 </div>
113 </div>
114 `).join('');
115 }
116
117 // ***** TOGGLE TEMPLATE BLOCK *****
118 // Opens/closes a collapsible section
119
120 function toggleBlock(i) {
121   const el = document.getElementById('tblock-' + i);
122   el.classList.toggle('open');
123 }
124
125 // ***** SET TODAY'S DATE *****
126
127 function setDate() {
128   const el = document.getElementById('today-date');
129   if (el) {
130     el.textContent = new Date().toLocaleDateString('en-IN', {
131       day: '2-digit', month: 'long', year: 'numeric'
132     });
133   }
134 }
135
136 // ***** INITIALISE *****
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```

11. Expected Output / Deliverables

The deliverable is a fully working front-end application with interactive activity tracking. Users can mark activities as completed and see real-time progress updates. The source code is organized into three files — `index.html`, `style.css`, and `script.js` — and uploaded to GitHub. The repository includes a proper folder structure and a `README.md` file that explains the project title, features, how to run it locally, and the technologies used. The application works correctly in any modern browser without installation.



The screenshot displays the 'FEWD Hackathon – Level 2' application interface. At the top, a header bar contains the logos for 'ORACLE UNIVERSITY' and 'ADROIT TECHNOLOGIES'. Below this, the title 'FEWD Hackathon – Level 2' is prominently displayed in a large, stylized font, with the subtitle 'Interactivity Implementation - Activity Tracker' underneath.

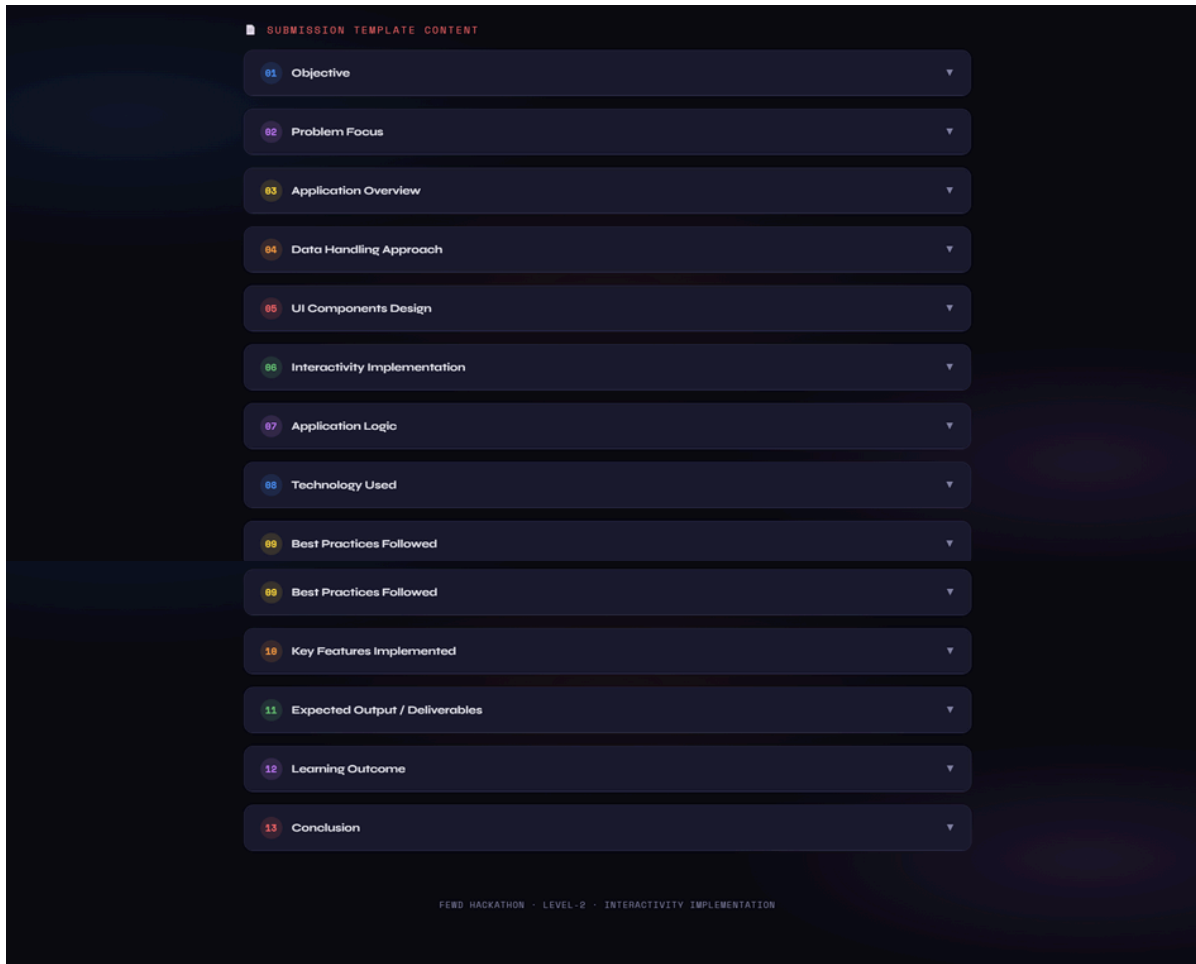
The main section is titled 'STUDENT DETAILS' and contains a form with the following fields:

- STUDENT NAME:
- YEAR:
- COLLEGE CODE:
- COLLEGE NAME:
- DEPARTMENT:
- SEMESTER:
- ROLL NO:
- NRID:
- DATE:
- TITLE:
- GITHUB LINK:

Below the student details, there is a 'LIVE PROGRESS' section showing '0 / 7 Completed' with a progress bar and the instruction 'Click any activity to mark it as completed.'

The bottom section is titled 'ACTIVITIES - CLICK TO TOGGLE' and lists seven activities, each with a description and a 'PENDING' status button:

- Set Up Project Structure**: Create HTML, CSS, JS files with clean folder layout.
- Design Activity List UI**: Style cards, badges, and layout with CSS.
- Load Activities from JS Array**: Dynamically render activity list from static data.
- Implement Click Event Handlers**: Attach event listeners to buttons / checkboxes.
- Update Status: Pending→Completed**: Toggle completed state and update DOM instantly.
- Real-Time Progress Counter**: Count completed, update progress bar & message.
- Push to GitHub with README**: Upload source with proper structure and documentation.



12. Learning Outcome

Through this project, the following skills were developed and strengthened: JavaScript event handling using onclick and addEventListener, DOM manipulation using getElementById, innerHTML, textContent, classList, and style, state management using plain JavaScript objects and boolean flags, dynamic UI rendering using template literals and array map(), progress calculation logic using filter() and Math.round(), handling edge cases in application logic, and applying CSS transitions and animations to create smooth, responsive visual feedback for user interactions.

13. Conclusion

This Level-2 submission successfully transforms the static Level-1 layout into a fully dynamic and interactive activity tracker. By applying JavaScript event handling, real-time DOM updates, and progress tracking logic, the application delivers a smooth and responsive user experience. Users can mark activities as completed and instantly see the progress bar and counter reflect their actions. The project demonstrates a clear understanding of core front-end interactivity concepts — event-driven programming, state management, and dynamic rendering — all implemented without any backend, framework, or page reload.